CS 202
Advanced Operating Systems

Winter 26

Lecture 6: Scheduling

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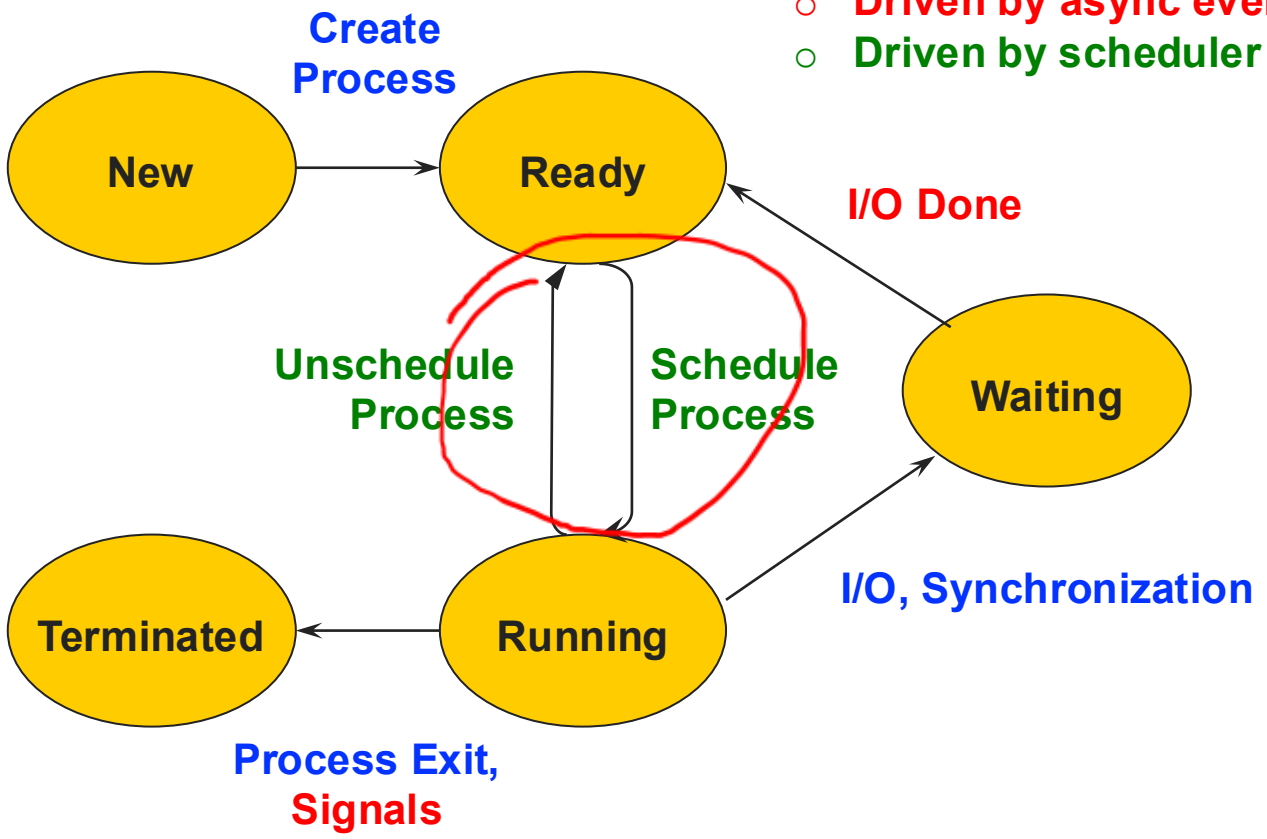
Execution State Graph

- Driven by process (system calls)
- Driven by async events
- Driven by scheduler



Multiprogramming

- Increase CPU utilization and job throughput by overlapping I/O and CPU activities
  - Doing this requires a combination of mechanisms and policy
- We have covered part of the mechanisms
  - Context switching, how and when it happens
  - Process queues and process states
  - Now meet the scheduler
- Now we'll look at the policies
  - Who, when, goals, …
- We'll refer to schedulable entities as jobs (standard usage) – could be processes, threads, people, etc.

Scheduling Overview

- Scheduler runs when we context switching among jobs to pick who runs next
  - Under what situation does this occur?
  - What should it do? Does it matter?
- Making this decision is called scheduling
- Now, we'll look at:
  - The goals of scheduling
  - Starvation
  - Various well-known scheduling algorithms
  - Standard Unix scheduling algorithm

Scheduler

- The scheduler is the OS module that manipulates the process queues, moving jobs to and from them
- The scheduling algorithm determines which jobs are chosen to run next, for how long, and what queues they wait on
- In general, the scheduler runs:
  - When a job switches from running to waiting
  - When an event occurs
    - » Why?
  - When a job is created or terminated
  - When a job voluntarily yield the CPU

Scheduling Levels

- Scheduling works at three levels in an operating system
  1. Control multiprogramming level – number of jobs loaded into memory
     - » Moving jobs to/from memory is often called swapping
     - » Long term scheduler: infrequent
  2. To decide where to run a job (multi-CPU, heterogeneous systems)
  3. To decide what job to run next
     - » Does it matter? What criteria?
     - » Short term scheduler: frequent
     - » We are concerned with this level of scheduling

Scheduling Styles

- Scheduler works differently in different systems
  - In preemptive systems the scheduler can interrupt a running job (involuntary context switch)
  - In non-preemptive systems, the scheduler waits for a running job to explicitly block (voluntary context switch)
  - What about preemptive kernel?
    - » Non-preemptive kernel disables maskable interrupts during event handling
    - » Preemptive kernel allows interrupts to be delivered during event handling → more responsive

Scheduling Goals

- What are some reasonable goals for a scheduler?
- Scheduling algorithms can have many different goals:
  - CPU utilization
  - Job throughput: # jobs/unit time
  - Turnaround time: $T_{finish} - T_{start}$
  - Avg Response time: avg time spent on ready queue
- Batch systems
  - Strive for job throughput, turnaround time (supercomputers)
- Interactive systems
  - Strive to minimize response time for interactive jobs (PC)
- Mobile systems
  - Strive to balance between performance and energy consumption

Starvation

Starvation is a scheduling “non-goal”:

- Starvation: a job is prevented from making progress because other jobs have the resource it requires
  - Resource could be the CPU, or a lock (later in synchronization)
- Starvation usually a side effect of the scheduling
  - E.g., a high priority process always prevents a low priority process from running on the CPU
- Starvation can be a side effect of synchronization
  - E.g., one thread always beats another when acquiring a lock
  - E.g., constant supply of readers always blocks out writers

Job Characteristics

- Achieving the goals may require knowing the job
  - Past
    - » When it arrived, how much progress has it made, how long has it run, has it been behaving nicely, …
  - Current
    - » How many resources it uses, how many are left, …
  - Future
    - » How much work is left? …
    - » Important for some scheduling algorithm, but can we really know?
  - Type
    - » GUI, I/O, realtime, …
  - Priority
  - etc

First In First Out (FIFO)

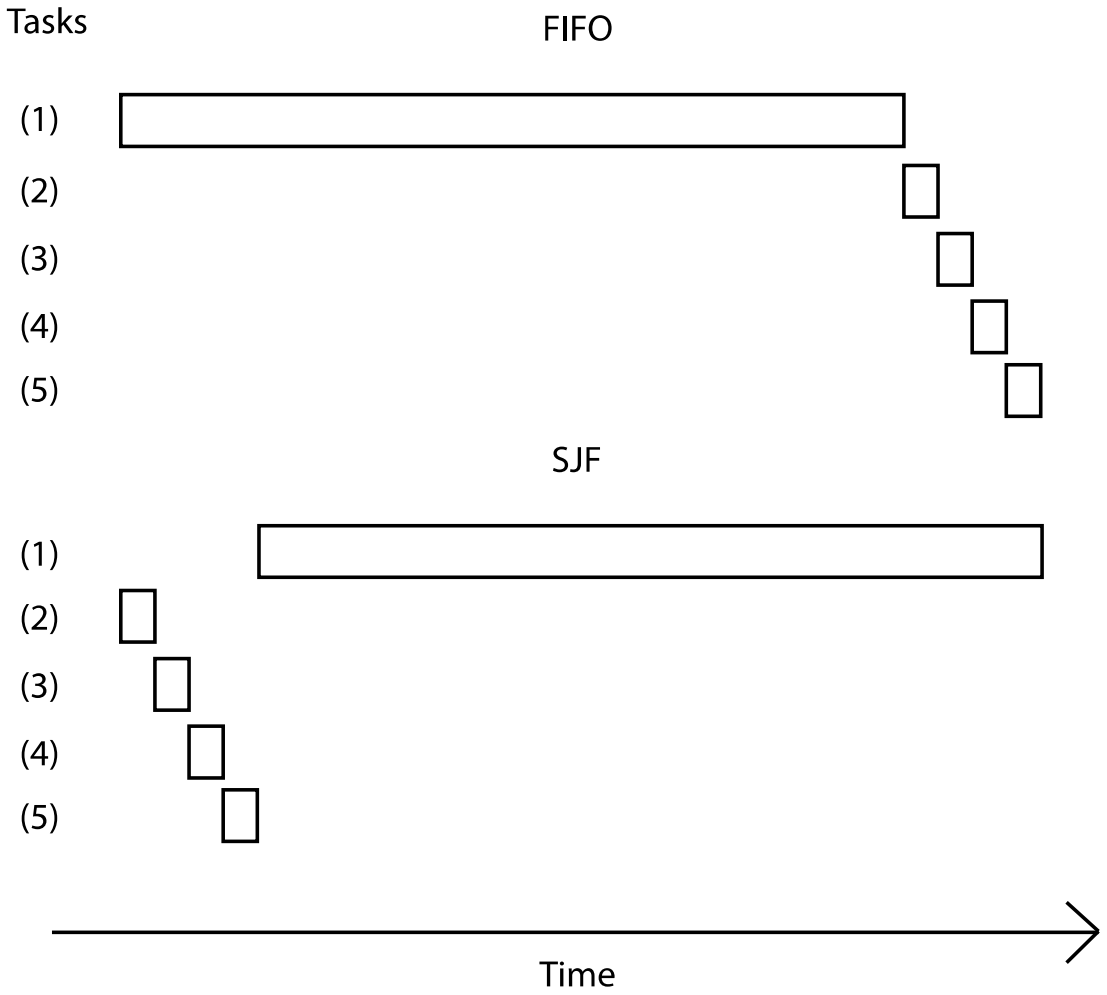
- Schedule tasks in the order they arrive
  - Continue running them until they complete or give up the processor
- Example: queues
  - Supermarket, banks, drive-through, …
- On what workloads is FIFO particularly bad?
  - Imagine being at supermarket to buy a drink of water but get stuck behind someone with a huge cart (or two!)
    - » …and who pays in pennies!
  - Can we do better?

Shortest Job First (SJF)

- Always do the task that has the shortest remaining amount of work to do
  - Often called Shortest Remaining Time First (SRTF)

- Suppose we have five tasks arrive one right after each other, but the first one is much longer than the others
  - Which completes first in FIFO? Next?
  - Which completes first in SJF? Next?

FIFO vs. SJF



What's the big deal? Don't they finish at the same time?

SJF

- Problems?
  - Impossible to know size of CPU burst
    - » Like choosing person in line without looking inside basket/cart
  - How can you make a reasonable guess?
  - Can potentially starve

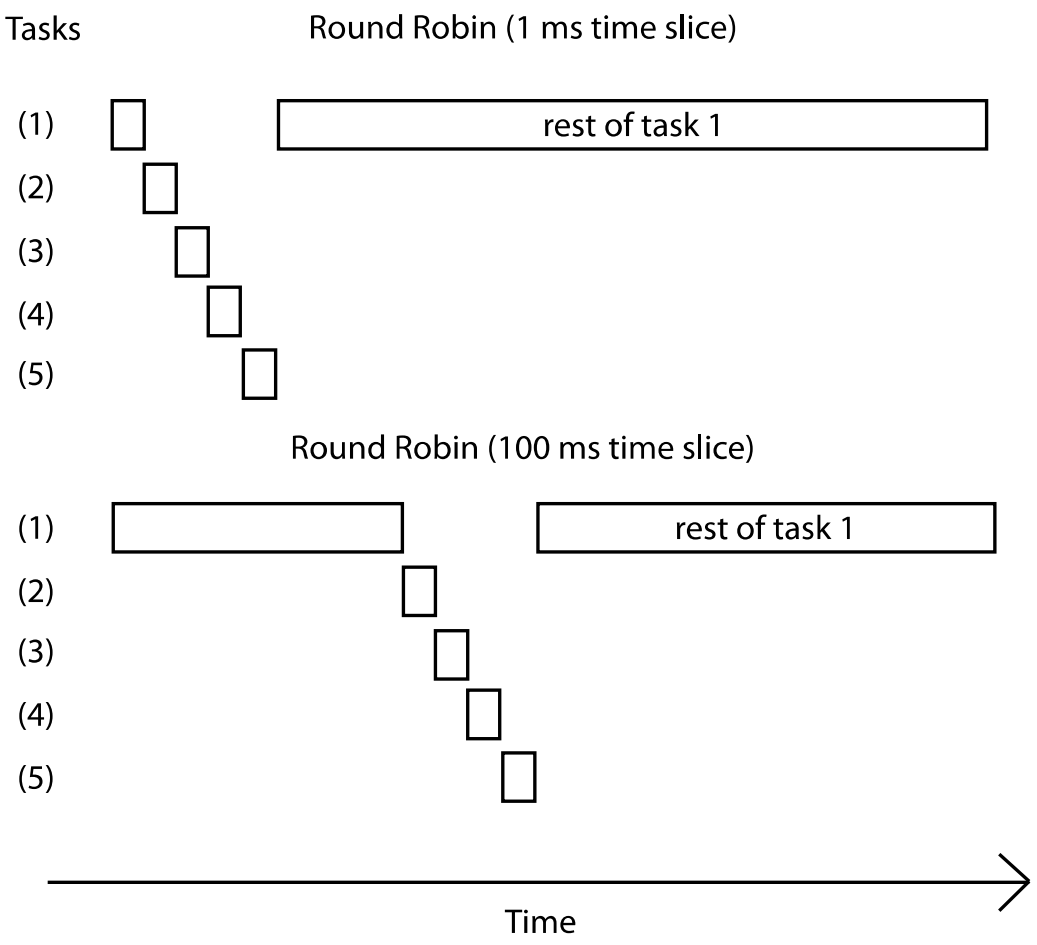
- Flavors
  - Can be either preemptive or non-preemptive
  - Preemptive SJF is called shortest remaining time first (SRTF)

Round Robin

- Each task gets resource for a fixed period (time quantum)
  - If task doesn't complete, it goes back in line

- Need to pick a time quantum
  - What if time quantum is too long?
    - » Infinite?
  - What if time quantum is too short?
    - » One instruction?

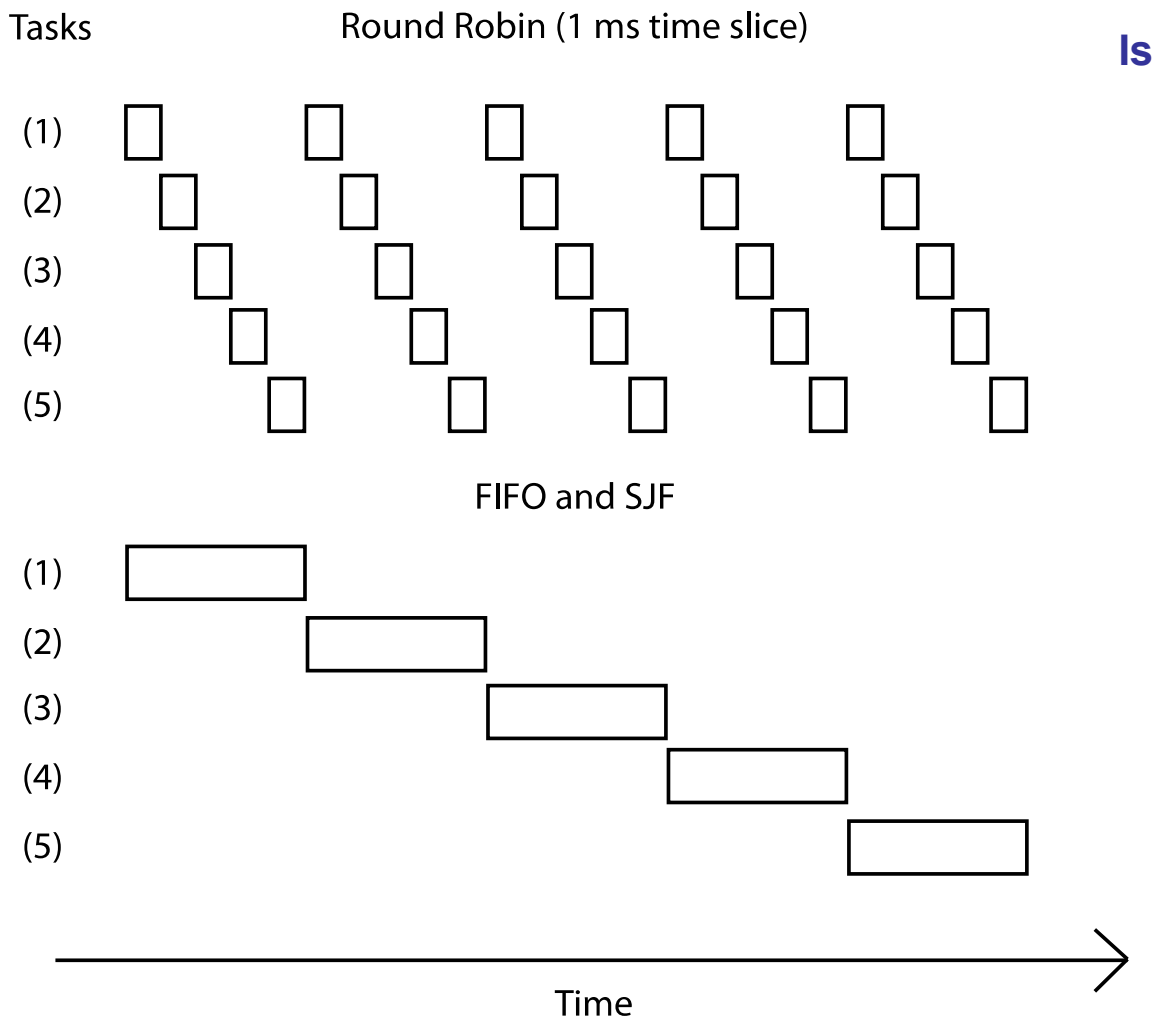
Round Robin


Tasks
Round Robin (1 ms time slice)
(1)
rest of task 1
(2)
(3)
(4)
(5)
Round Robin (100 ms time slice)
(1)
rest of task 1
(2)
(3)
(4)
(5)
Time

Round Robin vs. FIFO

- Many context switches can be costly
- Other than that, is Round Robin always better than FIFO?

Round Robin vs. FIFO



Is Round Robin always fair?

Problems of basic algorithms

- FIFO
  - Good: fairness; bad: turnaround time, response time
- SJF
  - Good: turnaround time, response time; bad: fairness, need to estimate run-time
- RR
  - Good: fairness, response time; bad: turnaround time
- Is there a scheduler that balances these issues better?
  - Challenge: limited information about a process in the beginning
  - Challenge: how to prevent gaming the scheduler to get more run-time

MLQ: combining algorithms

- Scheduling algorithms can be combined
  - Have multiple queues
  - Use a different algorithm for each queue
  - Move jobs among queues
- Example: Multiple-level feedback queues (MLFQ)
  - Multiple queues representing different job types
    - » Interactive, CPU-bound, batch, system, etc.
  - Queues have priorities, jobs on same queue scheduled RR
  - Jobs can move among queues based upon execution history
    - » Feedback: Switch from interactive to CPU-bound behavior

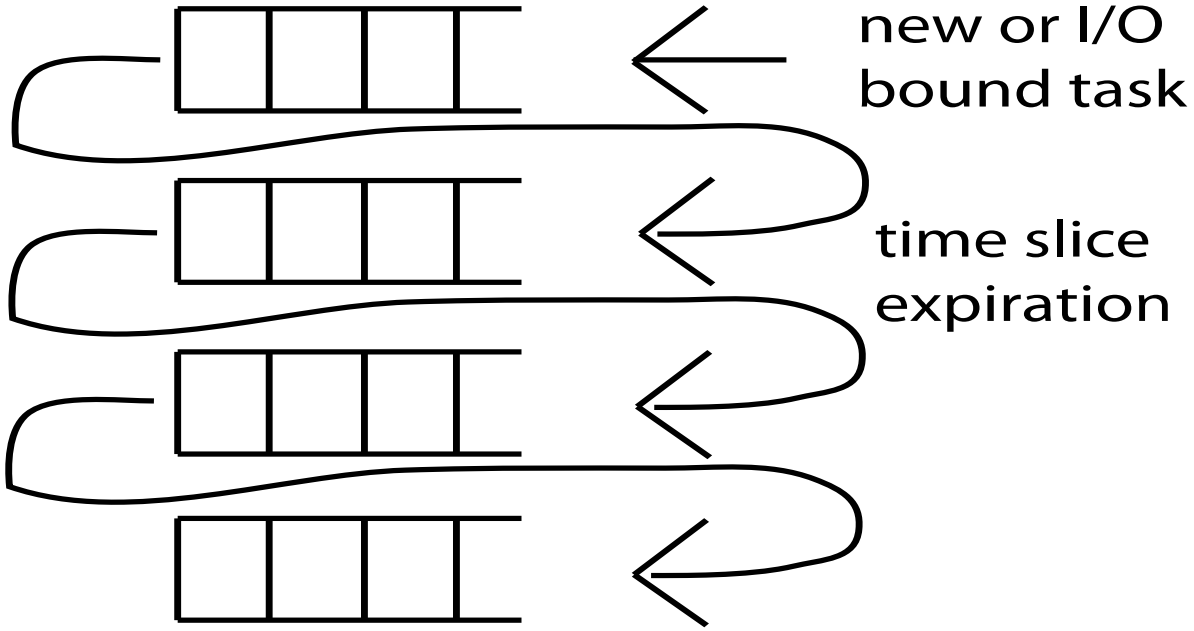
MLFQ

- Set of Round Robin queues
  - Each queue has a separate priority
- Higher priority queues have shorter time slices
  - Lower priority queues have longer time slices
  - Why?
- Scheduler picks first thread in highest priority queue
- Jobs start in highest priority queue
  - Assumes “good” behavior
- If a job used up the entire time slice, its priority drops one level
  - Otherwise it retains its priority

MLFQ

Priority	Time Slice (ms)
1	10
2	20
3	40
4	80

Round Robin Queues



MLFQ: Starvation and Gaming

- Wait … this design still allows starvation
  - Why?
- How to solve this issue?
  - After some period S, reset the priority by moving every job to the highest priority
- Can a job abuse the scheduler to increase its running time?
  - Jobs can deliberately relinquish the CPU before slice expires
  - Solution: using allotment time instead of one slice

Unix Scheduler

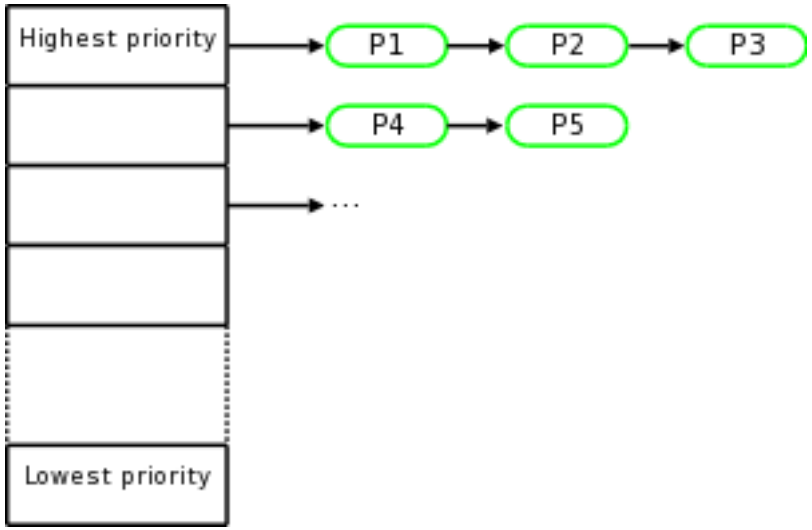
- The canonical Unix scheduler uses a MLFQ
  - 3-4 classes spanning ~170 priority levels
    - » Timesharing: first 60 priorities
    - » System: next 40 priorities
    - » Real-time: next 60 priorities
    - » Interrupt: next 10 (Solaris)
- Priority scheduling across queues, RR within a queue
  - The process with the highest priority always runs
  - Processes with the same priority are scheduled RR
- Processes dynamically change priority
  - Increases over time if process blocks before end of quantum
  - Decreases over time if process uses entire quantum

Motivation of Unix Scheduler

- The idea behind the Unix scheduler is to reward interactive processes over CPU hogs
- Interactive processes (shell, editor, etc.) typically run using short CPU bursts
  - They do not finish quantum before waiting for more input
- Want to minimize response time
  - Time from keystroke (putting process on ready queue) to executing keystroke handler (process running)
  - Don't want editor to wait until CPU hog finishes quantum
- This policy delays execution of CPU-bound jobs
  - But that's ok

The Rotating Staircase Deadline Scheduler*

- Based on MLQ, but with different rotation mechanism to provide better fairness and to prevent starvation
  - Priority Quota (minor rotation)
  - Active and Expired array (major rotation)



Lottery Scheduler

- Goal: fairness
  - Number of tickets represents the share of a resource that a process (or user or whatever) should receive
- Example: A (0 – 74), B (75 – 99)
  - Winning number: 63 85 70 39 76 17 29 41 36 39 10 99 68 83 63 62 43 0 49 12
  - Scheduling?
- Useful features
  - Ticket currency: Linux cgroup
  - Ticket inflation

Stride Scheduler

- Goal: fairness with better predictability
  - Stride: inverse of proportion of resources
  - Pass: accumulated “strides”
  - Scheduling: a sorted queue based on pass

Pass(A) (stride=100)	Pass(B) (stride=200)	Pass(C) (stride=40)	Who Runs?
0	0	0	A
100	0	0	B
100	200	0	C
100	200	40	C
100	200	80	C
100	200	120	A
200	200	120	C
200	200	160	C
200	200	200	...

The Completely Fair Scheduler

- A type of stride scheduler
  - Virtual runtime (vruntime): similar to pass, CFS schedule the process with the lowest vruntime
  - Using a red-black tree to maintain all runnable (i.e., in ready state) processes: O(log n) complexity
- Quota
  - Per CPU: sched latency / n (number of processes), minimal 6 ms
  - Weighting (niceness): when you’re too nice, you just don’t get as much (scheduling) attention
    - » Less nice (higher priority) processes get larger quota
    - » vruntime is increased proportionally too

Multiprocessor Scheduling*

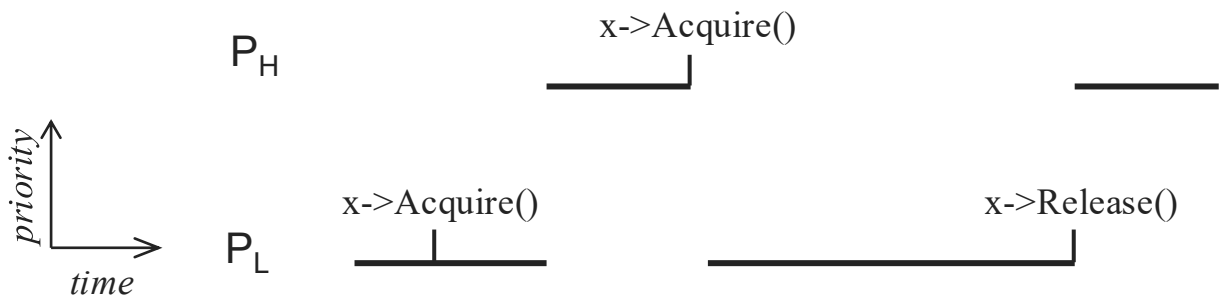
- This is its own topic, we won't go into it in detail
  - Could come back to it towards the end of the quarter
- What would happen if we used MLFQ on a multiprocessor?
  - Contention for scheduler spinlock
  - Multiple MFQ used – this optimization technique is called distributed locking and is common in concurrent programming
- A couple of other considerations
  - Co-scheduling for parallel programs
  - Core affinity

Scheduling Summary

- Scheduler (dispatcher) is the module that gets invoked when a context switch needs to happen
- Scheduling algorithm determines which job runs next, where jobs are placed on queues
- Many potential goals of scheduling algorithms
  - Utilization, throughput, wait time, response time, fairness, etc.
- Various algorithms to meet these goals
  - FCFS/FIFO, SJF, RR, Priority
- Can combine algorithms
  - Multiple-level feedback queues
  - Unix example

More on Priority Scheduling

- For real-time (predictable) systems, priority is often used to isolate a process from those with lower priority. *Priority inversion* is a risk unless all resources are jointly scheduled.



Priority Inheritance

- If lower priority process is being waited on by a higher priority process it inherits its priority
  - How does this help?
  - Does it prevent the previous problem?

- Priority inversion is a big problem for real-time systems
  - Mars pathfinder bug (link)